Supplementary Material for:
Networks Decide for Us: Emergence of Adoption in Homophily-Imputed Social Topologies

Aníbal Olivera Morales, Jorge Fábrega Lacoa

March 27, 2026

Contents

S1 Survey Items and Propensity Distributions

To assign behavioral propensities (Minimum Utility Requirements, q_i) to the agents in our imputed topologies, we utilized specific batteries of items from the American Trends Panel (ATP).

S1.1 Openness to Innovation Score (ATP 2014)

The openness to innovation score was constructed using binary responses to the following items:

1. Usually try new products before others do. (*Pro-innovation*)
2. Prefer my tried and trusted brands. (*Anti-innovation*)
3. Like being able to tell others about new brands and products I have tried. (*Pro-innovation*)
4. Like the variety of trying new products. (*Pro-innovation*)
5. Feel more comfortable using familiar brands and products. (*Anti-innovation*)
6. Wait until I hear about others' experiences before I try new products. (*Anti-innovation*)

S1.2 Collective Action Propensity Score (ATP 2016)

For collective action simulations, the propensity score was generated from responses regarding political and social participation. Respondents indicated whether they had engaged in actions (e.g., in the past year, distant past, might do it, or would never do it):

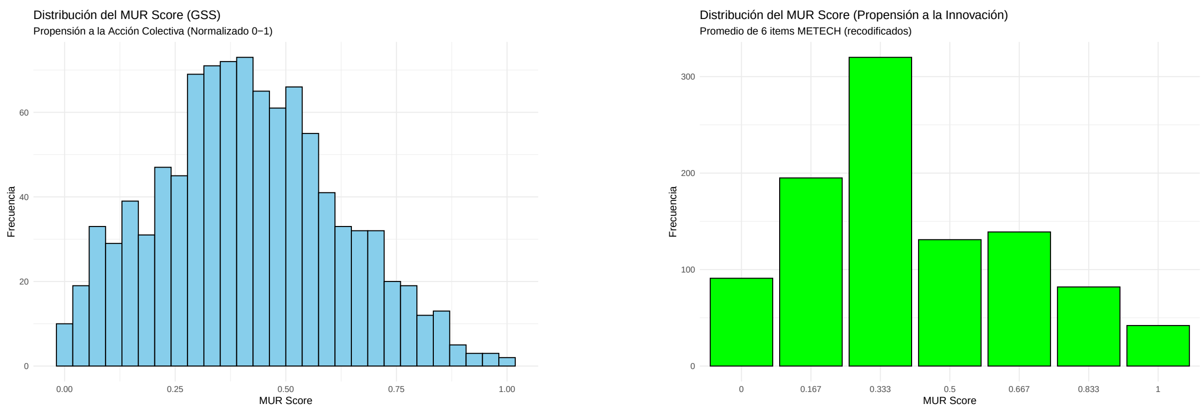
1. Signed a petition.
2. Boycotted, or deliberately bought, certain products.
3. Took part in a demonstration.
4. Attended a political meeting or rally.
5. Contacted a politician or a civil servant to express views.
6. Donated money or raised funds for a social/political activity.
7. Contacted or appeared in the media to express views.
8. Joined an Internet political forum or discussion group.

S1.3 Collective-Action Propensity: Construct Internal Reliability

To ensure the empirical plausibility of the ATP mapping, we derive a unidimensional Minimum Utility Requirement (MUR) score. [PLACEHOLDER: Add Cronbach's alpha reliability test to verify multi-item unidimensionality for the collective-action propensity construct].

S1.4 Empirical Distribution of the Intrinsic Utility Level (IUL) / MUR

The following distribution histograms (Figure S1) reflect the normalized scores extracted from the ATP and GSS micro-data. We leverage the GSS structure for Figure 1 in the main manuscript because its survey distribution allows for a more granular, finely binned *collective-action propensity score*. This expanded level-count avoids artifactual stepping and renders smooth analytical heatmaps capable of correctly tracing criticality variance signatures across contiguous thresholds.



(a) MUR Profile (GSS 2004).

(b) MUR Profile (ATP 2014/2016).

Figure S1: Distribution of Minimum Utility Requirements (q_i). Lower MUR implies higher propensity towards innovation.

S1.5 Criticality Heatmaps in Randomized (Erdős-Rényi) Networks

For direct comparison with the main structural signatures found across Plausible domains in the main text, Figure S2 documents the equivalent heatmaps corresponding to an Erdős-Rényi graph. This baseline identically preserves total node degree and density, but destroys local multidimensional homophily and structural clustering.

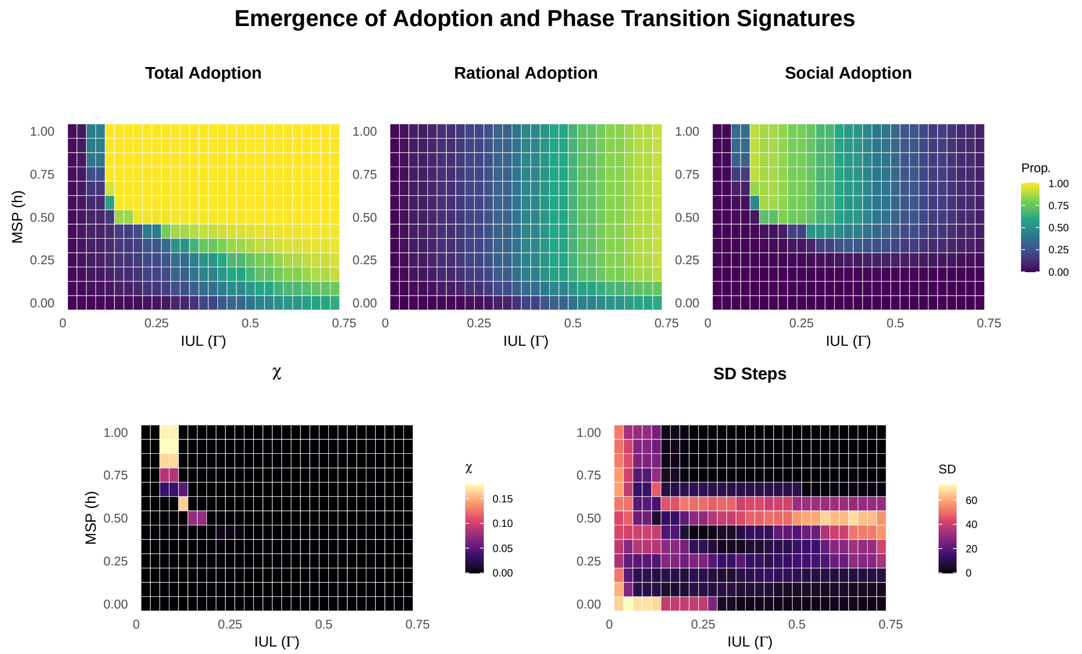


Figure S2: Phase Transition Signatures under generic Erdős-Rényi baseline. Note the complete collapse of variance boundaries and the instantaneous system-wide discontinuity, severing the stable transition regimes found in empirically clustered homophilic topologies.

S2 Agent-Based Model: ODD Protocol Summary

To ensure reproducibility, we outline the primary rules of our complex contagion model following the Overview, Design concepts, and Details (ODD) protocol logic.

Initialization (Seeding) Depending on the experimental block, the simulation initializes with either a random node or a strategically pivotal node (e.g., highest Degree, Closeness, Eigenvector). If the selected node requires social reinforcement (threshold count > 1) to adopt, the algorithm forces adoption on the initiator and iteratively seeds up to the required threshold boundary of the initiator's neighborhood.

Updating (Stochastic Exposure) Agent behavior is updated using a stochastic, asynchronous network exposure model via `netdiffuseR`. At each simulation tick, non-adopters calculate their exposure to adoptions in their local neighborhood. Instead of deterministic synchronous updates—which often trap systems in artificial, oscillating stable states—agents probabilistically transition based on their specific threshold profiles τ_i, which are drawn from $\sim N(\mu, \sigma^2)$ distributions spanning rational and social components.

Stopping Criteria The simulation halts under two conditions: (1) an absorbing state is reached wherein no novel adoptions occurred during the current iteration (verified internally by the non-changing diffusion state), or (2) the simulation exceeds a maximum execution limit of $t = 200$ steps. Throughout our parameter sweeps, $t = 200$ safely encapsulated the full cascade lifetime even at critical phase transitions.

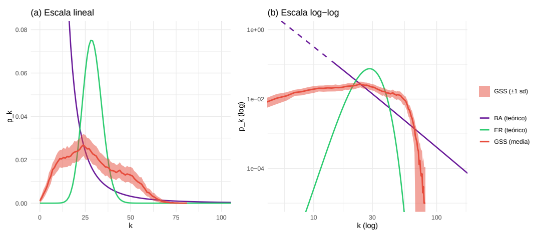
S3 Network Imputation and Topological Validation

Following McPherson and Smith (2019), we mapped ego-centric data from the General Social Survey (GSS 2004) onto the ATP sample to construct whole networks ($N = 1000$) using Exponential Random Graph Models (ERGMs).

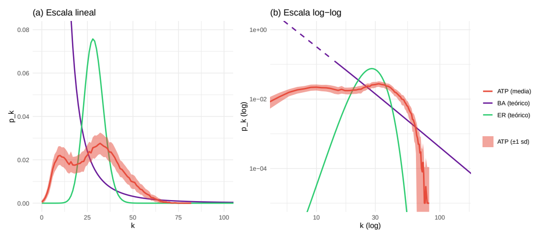
The reliance on empirical topologies instead of synthetic baselines is theoretically motivated: as Talaga and Nowak (2020) note, "real-world social networks often exhibit a set of characteristic structural properties that do not occur jointly as often in different kinds of complex networks ... well-known generic models such as Erdős-Rényi random graphs or preferential attachment networks may not fit well to the social problem at hand."

To reflect this reality, our ERGM directly incorporated fixed homophily coefficients mapping multiple dimensions in Blau space: categorical homophily for race (`nodematch("race")`), sex, and religion, alongside continuous absolute difference effects for age (`absdiff("age")`) and years of education. The base likelihood for forming an edge (`edges`) was calibrated iteratively via MCMC simulations (burn-in $= 1000 \times N$, interval $= 100 \times N$) until reaching a stable expected empirical target density of ≈ 0.029. We generated 100 independent topological realizations to act as our Plausible simulation substrates.

Figure S3 demonstrates that the generated pseudo-empirical networks successfully maintain realistic, right-skewed topological features that traditional models fail to capture jointly. Furthermore, Figure S4 validates that empirical structures sustain high localized clustering across degree scales compared to theoretical controls.

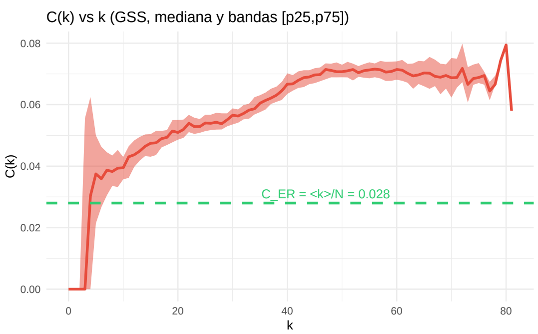


(a) GSS respondents.

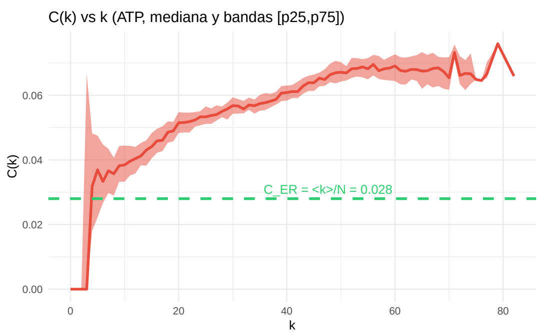


(b) ATP respondents.

Figure S3: Degree distributions of the imputed networks plotted against theoretical models (Degree-Preserving Randomized, Barabási-Albert). The empirical imputations accurately capture real-world right-skewed degree structures.



(a) Clustering vs. Degree (GSS).



(b) Clustering vs. Degree (ATP).

Figure S4: Local clustering coefficient $C(k)$ as a function of degree k. The imputed topologies maintain rich local triad closures characteristic of social structures.

S4 Extended Criticality Analysis: Plausible vs. Randomized

The mathematical relevance of recovering the critical exponent $\gamma = 1$ in the main text is rooted in Landau's phenomenological theory of phase transitions.

S4.1 Mean-Field Derivation

The free energy density $f(\Phi)$ of the interacting system expanded around the order parameter Φ is:

$$f(\Phi) = f_0 + \frac{1}{2}a(T - T_c)\Phi^2 + \frac{1}{4}b\Phi^4 - H\Phi \tag{S1}$$

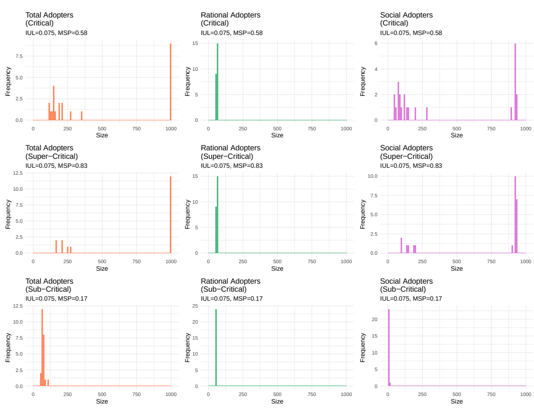
Minimizing free energy requires $\frac{\partial f}{\partial \Phi} = 0$. By differentiating the equilibrium condition with respect to an external field H, we find the isothermal susceptibility χ:

$$a(T - T_c)\chi + 3b\Phi^2\chi - 1 = 0 \implies \chi = \frac{1}{a(T - T_c) + 3b\Phi^2} \tag{S2}$$

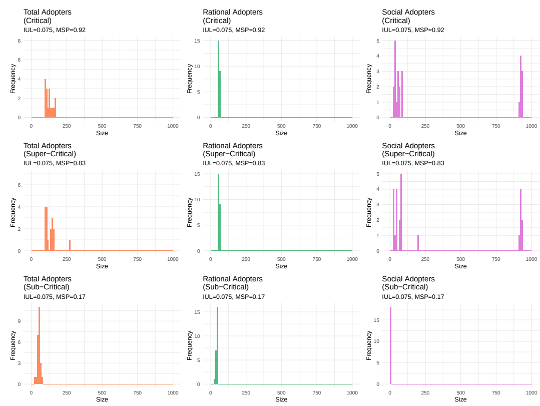
As $H \to 0$, $\Phi \to 0$ for $T > T_c$. Thus, susceptibility diverges strictly as $\chi \sim |T - T_c|^{-1}$. Extracting an empirical $\gamma \approx 1$ confirms the system obeys continuous macroscopic scaling.

S4.2 Avalanche Size Spectra and Critical Scaling

As shown below, empirical GSS topologies exhibit scale-free power-law bimodality at criticality, indicative of a second-order transition. The Erdős-Rényi random graphs exhibit collapsed binary spectra, representing a first-order discontinuity. To rigorously trace these transitions, we sweep the simulation domain while fixing the Intrinsic Utility Level (IUL) constant. As the systemic threshold (MSP) crosses the empirical boundary MSP_c, cascading "avalanches" (adoption sizes) transition structurally.

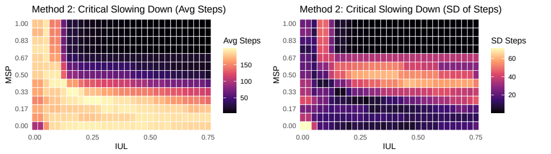


(a) GSS: Scale-free power-law bimodality.

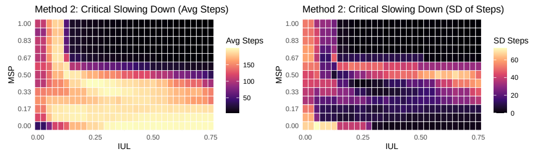


(b) Erdős-Rényi: Collapsed binary cascade spectra.

Figure S5: Avalanche Size Spectra comparison evaluated around the critical variance boundaries $\max(\mathrm{Var}(\Phi))$.



(a) GSS: Divergence indicates continuous domains.



(b) Erdős-Rényi: Suppressed transition boundaries.

Figure S6: Critical Slowing Down: Variance of required execution ticks before the structural plateau.

S5 BAM Regression Model Surfaces

To ensure the robustness of the structural premium, we explored interacting non-linear terms across multiple dimensions. The following figures visualize the non-linear interaction terms $s(\mathrm{IUL}, \mathrm{MSP})$ estimated by the Big Additive Models (BAMs) across all contagion types (Innovation and Collective Action) for Rational, Social, and Total Adoption.

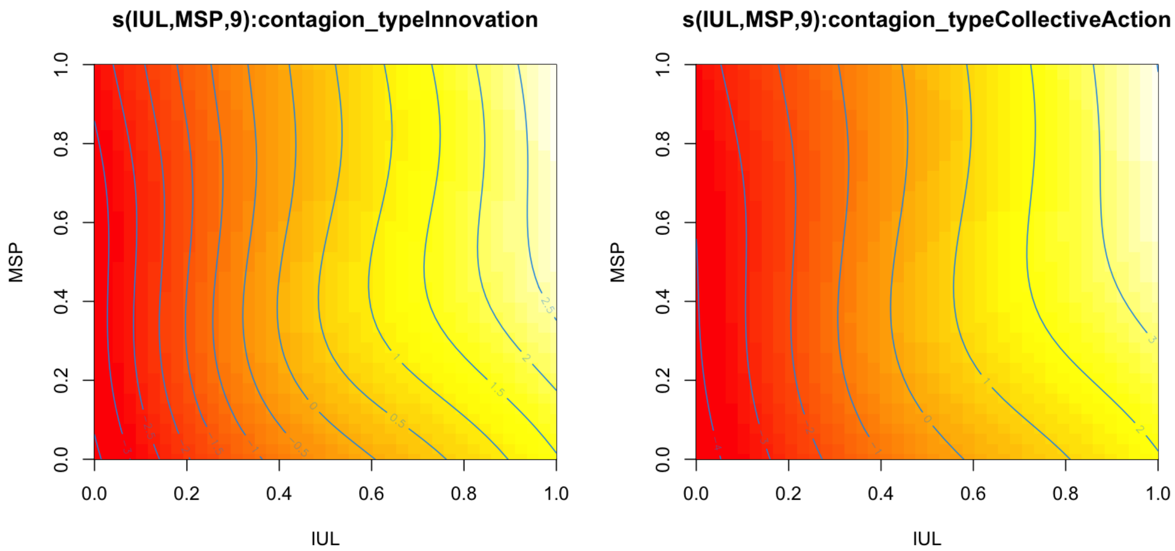


Figure S7: BAM Smooth Surfaces for Rational Adoption.

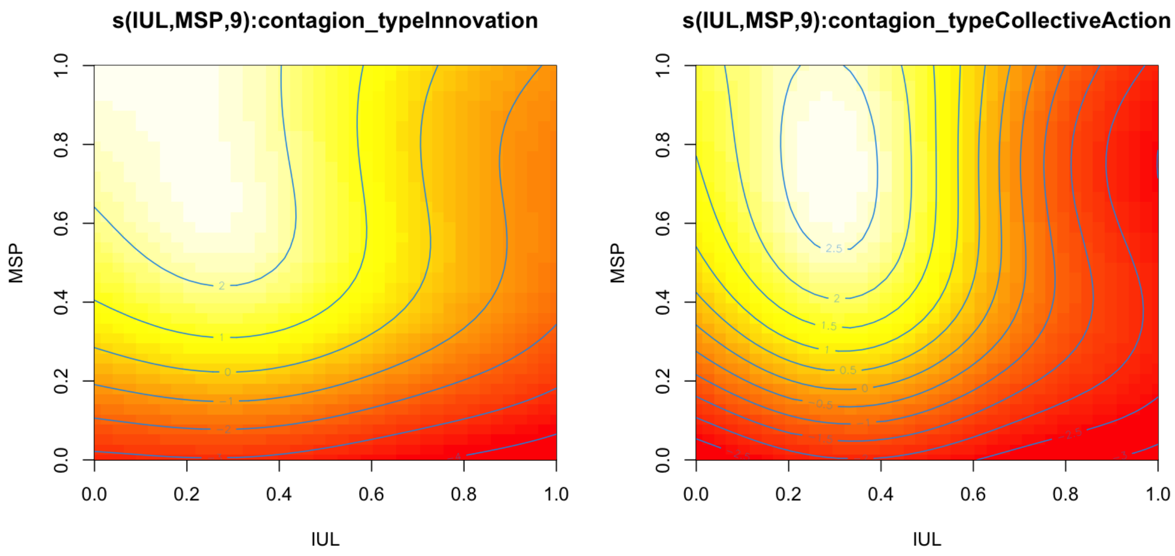


Figure S8: BAM Smooth Surfaces for Social Adoption.

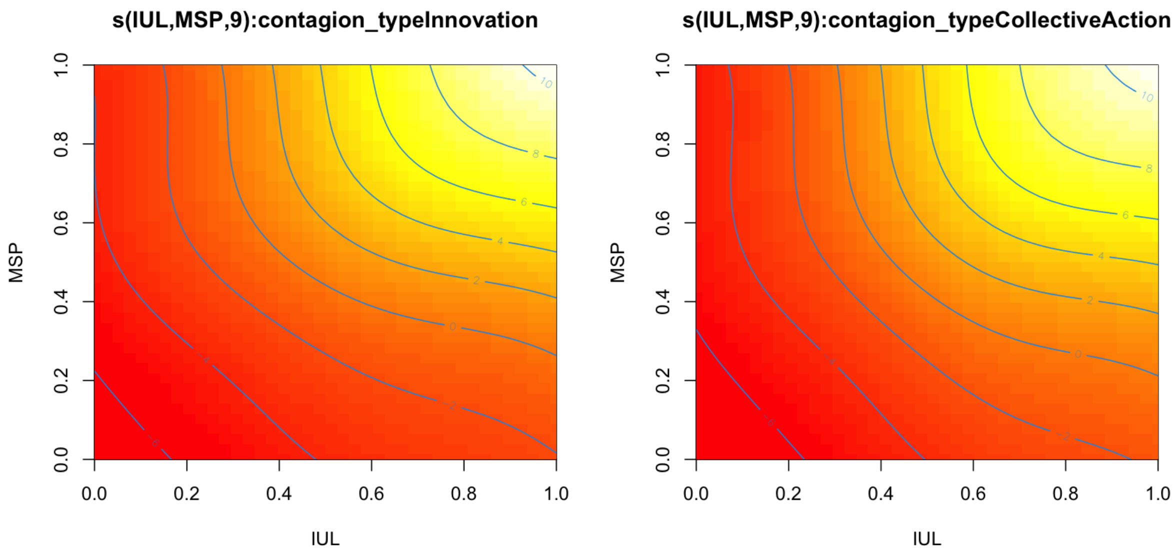


Figure S9: BAM Smooth Surfaces for Total Adoption.